

MOHAMMAD BILAL

PROFESSIONAL SUMMARY

Software Engineer with experience in Python backend development, cloud services, full-stack applications, and machine learning solutions. Skilled in building REST APIs, secure web applications, automated delivery workflows, and AWS-based infrastructure. Strong problem-solving and communication abilities supported by hands-on development and debugging.

TECHNICAL SKILLS

Languages	Python, TypeScript, Java, JavaScript, SQL
Backend	Django, Django REST Framework, Flask, Spring Boot, Node.js, REST APIs
Frontend	React, Next.js, HTML, Tailwind CSS
Cloud & DevOps	AWS, Docker, Terraform, GitHub Actions, Nginx, Linux
AWS	Lambda, S3, EC2, IAM, API Gateway, CloudWatch,
Data & Storage	PostgreSQL, MySQL, MongoDB, SQLite, Redis, Prisma
AI & Hardware	Machine Learning, OpenCV, YOLOv4, Data Science, Raspberry Pi

PROFESSIONAL EXPERIENCE

LTM | Backend Engineer

- Developing backend, cloud, and full-stack engineering capabilities through hands-on applications using Python, TypeScript, AWS, Docker, and infrastructure automation.
- Built production-oriented solutions involving REST APIs, authentication, databases, testing, CI/CD, containerization, and cloud deployment.

Quadvare | Python Developer

- Contributed to a machine-learning-based voice automation solution that helped patients book hospital appointments through phone calls.
- Supported Python backend development and conversational booking workflows.

PROJECTS

E-Commerce Platform | Next.js, TypeScript, Django REST Framework, PostgreSQL, AWS, Docker

- Developed an end-to-end platform for catalog management, inventory, guest checkout, payments, shipping, order tracking, and role-based administration.
- Designed REST APIs, data models, RBAC, asynchronous workflows, automated testing, CI/CD, and an AWS deployment strategy.

Online Academy Management Platform | Next.js, React, TypeScript, Prisma, Docker, Terraform, AWS

- Built a tutoring platform with service pages, demo booking, verified testimonials, role-based authentication, and an admin moderation dashboard.
- Implemented APIs, validation, security controls, automated tests, containerization, and infrastructure configuration.

Lane and Object Detection System | Python, YOLOv4, OpenCV

- Developed a real-time computer vision pipeline for lane and vehicle detection from live and recorded video streams.

Book Selling Platform | Python, Flask, MongoDB

- Created a web application with product listings, seller authentication, customer ordering, and real-time inventory updates.

AI-Powered Companion Robot | Python, Raspberry Pi, IoT

- Engineered a Raspberry Pi assistant with alarms, note-taking, IoT device control, and an interactive visual interface.

ADDITIONAL EXPERIENCE

Freelance Online Tutor | Mathematics & Programming

- Teach school students through interactive, problem-oriented sessions focused on logical thinking, mathematics, and programming.

EDUCATION

Bachelor of Technology in Computer Science and Engineering | Specialization in Data Science

SCMS School of Engineering and Technology, APJ Abdul Kalam Technological University

CERTIFICATIONS

AWS Certified AI Practitioner • AWS Certified Cloud Practitioner • Microsoft Certified: Azure AI Fundamentals

ACHIEVEMENTS & DESIGN REGISTRATION

- Secured 1st place in Kerala and Global Rank 226 in the IEEE Xtreme Programming Competition.
- Registered the design for the Lifestyle Utility Monitoring and Interface (LUMI) Device with the Indian Designs Registry, Design No. 462137-001.
- Selected for the EBSB Yuva Sangam student exchange initiative by the Government of India.